

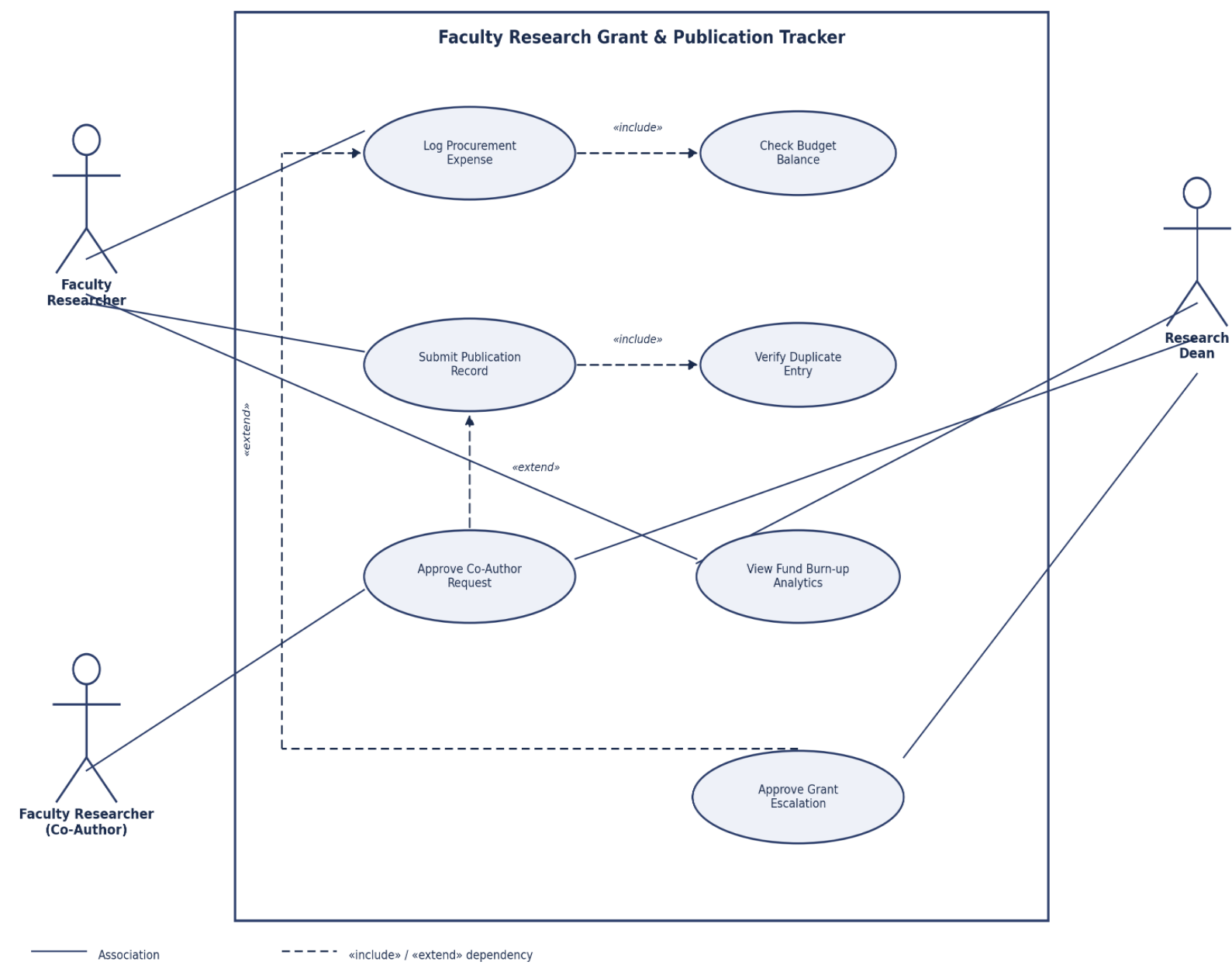
Faculty Research Grant & Publication Tracker

Complete Requirements Table — Problem Statement #09

Functional Requirements						
	ID	Description	Priority	Acceptance Criteria	Rationale	
	FR-001	The system shall track grant fund allocations, allow faculty to log equipment/consumable procurement expenses against a grant, and compute the remaining budgetary balance in real time.	High	Pass: Balance updates accurately after each expense and the system blocks any expense that would exceed the grant cap unless an admin override is applied. Fail: Overdraft is allowed without an admin override, or the displayed balance does not match the sum of allocations minus expenses.	Prevents grant overdrafts and gives faculty real-time visibility into spendable funds, which is the core financial control the research deanery needs.	
	FR-002	The system shall allow faculty to submit publication records including title, authors, venue, DOI, indexing status (Scopus/WoS/UGC-CARE), and citation count, and shall check each submission for duplicate DOI/title entries before acceptance.	High	Pass: A duplicate DOI or high-similarity title is detected and the submission is blocked with a clear conflict message; unique submissions are stored successfully. Fail: A duplicate record is silently created, or a valid unique submission is rejected.	Accurate, de-duplicated publication data is essential for correct citation/indexing metrics reported to the deanery and external ranking bodies.	
	FR-003	For publications with more than one author, the system shall route a co-author approval request to each declared co-author and shall only mark the publication as “verified” once every co-author has approved the authorship order and contribution share.	Medium	Pass: Publication status changes to “verified” only after 100% of co-authors approve; a single rejection reverts the record to draft. Fail: A publication is marked verified while a co-author approval is still pending or rejected.	Protects academic integrity and ensures all contributing authors have consented to how credit is recorded before it counts toward departmental metrics.	
	FR-004	The system shall generate fund burn-up analytics, visually plotting cumulative grant expenditure against sanctioned budget and time-based milestones, filterable by grant, category, and date range.	Medium	Pass: The burn-up chart renders correct cumulative totals matching the underlying ledger for any selected filter combination. Fail: Chart totals diverge from the ledger, or filters produce no data / incorrect date ranges.	Gives the Research Dean an at-a-glance view of spending velocity across grants, supporting proactive budget management ahead of grant closure deadlines.	
	FR-005	The system shall automatically notify the Research Dean whenever a grant's cumulative fund utilization crosses the 75%, 90%, or 100% threshold, and shall require Dean approval before further expenses can be logged past the breached threshold.	High	Pass: A notification is generated within the same transaction that crosses a threshold, and further expense logging is blocked until the Dean approves or rejects escalation. Fail: Expenses continue to be logged past a threshold with no notification or approval gate.	Provides an early-warning and control mechanism so grants are not exhausted or overdrawn without deanery visibility and sign-off.	
Non-Functional Requirements						
	ID	Type	Description	Priority	Acceptance Criteria	Rationale
	NFR-001	Security & Auditability	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals (expense logging, escalation decisions) and journal/publication status modifications, recording actor, timestamp, and before/after state.	High	Pass: Benchmarking and penetration tests confirm every financial/status-change event produces a cryptographically verifiable ledger entry that cannot be edited or deleted, under simulated peak load. Fail: Any ledger entry can be silently altered, or an approval/status change occurs without a corresponding entry.	Sponsored research funds and academic records require a defensible, tamper-proof trail for internal audits and external funding-agency compliance reviews.

NFR-002	Availability & Scalability	The system shall be available with at least 99.5% uptime during active grant-submission and publication-reporting windows, and shall support at least 500 concurrent authenticated users without degradation of response time beyond 2 seconds for standard read operations.	Medium	Pass: Load testing at 500 concurrent users maintains sub-2-second response for standard reads and uptime monitoring shows ≥99.5% availability over a rolling 30-day window. Fail: Response times exceed target under load, or uptime falls below 99.5% during a reporting window.	The portal must remain responsive during peak periods such as grant-renewal deadlines and annual publication reporting cycles, when faculty and deanery usage spikes simultaneously.
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UML USE CASE DIAGRAM:



Brief Description

Covers the optional co-author approval workflow that is only triggered when a publication submission includes a co-author who must confirm authorship and attribution share before the publication record is finalized. This is an extension of Submit Publication Record, invoked only under a specific condition.

Preconditions

- | A publication record exists in “pending verification” status with one or more declared co-authors.
- | The extension point “Add Co-Author” was reached during Submit Publication Record because co-authors were specified.

Postconditions

- | Each named co-author's approval status is recorded as Approved or Rejected.
- | If all co-authors approve, the publication record status changes to “verified.”
- | If any co-author rejects, the record is returned to “draft” status for correction.

Main Success Scenario

- 1 System (extending Submit Publication Record at the “Add Co-Author” extension point) sends an approval request to each declared co-author.
- 2 Co-Author Faculty Researcher reviews the authorship order and contribution percentage.
- 3 Co-Author approves the attribution via the portal.
- 4 System records the approval and checks whether all co-authors have responded.
- 5 Once all co-authors have approved, System marks the publication record as “verified” and notifies the Research Dean.

6 Research Dean reviews the verified record and confirms it for inclusion in departmental metrics.

Alternate Flow — Co-Author Rejects Attribution

1 At step 3, a Co-Author disputes the listed contribution percentage or authorship order and selects “Reject.”

2 System records the rejection reason and reverts the publication record to “draft” status.

3 System notifies the original submitting Faculty Researcher of the rejection and reason.

4 Submitting Faculty Researcher edits the authorship details and re-triggers the approval workflow, resuming at step 1.

Exception / Error Flow

If a co-author does not respond within the configured SLA window (e.g., 10 working days), the system escalates the pending approval to the Research Dean as a reminder notification, without altering the record status.

Notes / UML Relationship Detail

Modeled with an «extend» relationship (not «include») because Approve Co-Author Request runs only conditionally — when co-authors are present at submission time — and it inserts optional behavior at a named extension point rather than being mandatory on every execution of the base use case